

Who will charge?

Level 1 — Binary Classification Task

Crack the pattern. Predict the plug. ⚡

Background

⚡ E-mobility is growing rapidly — and so is the complexity behind it. Millions of charging sessions generate data every day, and that data is critical for one thing: accurate load forecasts that keep charging infrastructure, buildings, and power grids stable.

In this challenge you will work with **real, unfiltered data from genuine KEBA charging sessions** and develop AI models for accurate load forecasting. Intelligent forecasting is a key lever in modern energy systems: your model could directly influence how real-world charging infrastructure is designed, operated, and scaled. No pressure. 😄

Dataset

You receive a **time series** of electric vehicle (EV) charging sessions collected from real KEBA charging hardware. Each row in the dataset corresponds to one timestep of the time series.

Input	Time series of EV charging session data (CSV format)
Target column	Binary indicator — 1 if a user is currently charging, 0 otherwise
Task type	Binary Classification

The input file contains timestamped features derived from historical charging behaviour. Your model must output a **probability** (a value between 0 and 1) for each row in the test set, indicating the likelihood that a user is actively charging at that timestep.

Your Task

Train a binary classification model that predicts, for each timestep, **whether a user is currently charging** (target = 1) or not (target = 0). 🔌

You are free to choose any machine learning approach — classical statistics, gradient boosting, neural networks, or whatever you have up your sleeve. What matters is the quality of your probabilistic predictions on the hold-out test set. Time to flex those model weights. 💪

Submission Format

Submit a single **CSV file** via the contest platform. The file must contain:

- A column labeled **unique_id** that contains the **id** of the user — same as in `submission_example_level_1.csv`
- A column labeled **ds** that contains the hourly datetime — same as in `submission_example_level_1.csv`
- A column labeled **y_prob** with **predicted probabilities** (floats in [0, 1]) indicating the likelihood of an active charging session at that timestep

Optionally, you may also upload your source code for transparency and reproducibility.

Evaluation Metrics

Your submission is evaluated against three complementary metrics 📊 — Brier Score, Log Loss, and ROC-AUC. Only the **Brier Score** is visible to you on the platform; the other two are computed internally and do not appear on the leaderboard.

Brier Score (BS)

Mean squared error between predicted probabilities and true binary labels. A perfectly calibrated model achieves BS = 0; a model that always predicts 0.5 scores 0.25. Lower is better.

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2$$

p_i = predicted probability for sample i, $y_i \in \{0, 1\}$ = true label, N = number of samples

ROC-AUC (AUC)

Area under the Receiver Operating Characteristic curve. Measures rank-ordering quality across all decision thresholds — i.e. how reliably your model assigns higher probabilities to true positives. Random baseline = 0.5; perfect = 1.0. Higher is better.

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}(\mathrm{FPR})\, d(\mathrm{FPR})$$

TPR = True Positive Rate, FPR = False Positive Rate

Log Loss (LL)

Negative log-likelihood of the predicted probability distribution. Penalises overconfident incorrect predictions super-linearly — the closer a wrong prediction is to certainty, the steeper the penalty. Perfect calibration → LL = 0. Lower is better.

$$\mathrm{LL} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log p_i + (1-y_i)\log(1-p_i)]$$

Predictions clipped to avoid $\log(0)$. p_i = predicted probability, y_i = true label

🏆 Final Rank

The leaderboard rank for Level 1 is determined by **Brier Score** — the lower, the better. This is the only metric visible to you on the platform. Log Loss and ROC-AUC are computed internally for analysis purposes only.

⚡ Plug in, power up, and let your models run — we're excited to see what you build. Good luck! 🤖

📧 Questions, platform issues, or feedback? Don't hesitate to reach out — ai-contest@keba.com

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